VORLDX INDUSTRIES PRIVATE LIMITED • CIN: U62099UP2026PTC245598

VorldX Employee Attendance & Workforce System

Comprehensive Technical Project Report & Managerial Architecture Review

Document Version	Release Date	Evaluation Scope	Production Status
2.1.0 (Production Verified)	October 31, 2026	Full-Stack Web, Mobile, Kiosk & API	🟢 100% Live & Verified (Zero Regression)

1. Executive Summary & Business Problem Statement

Legacy corporate attendance tracking mechanisms (biometric fingerprint readers, plastic RFID badges, and static QR codes) impose steep financial, legal, and operational risks in modern hybrid work environments:

- **Buddy Punching & Proxy Attendance:** Static QR codes or shared links can be photographed, forwarded across instant messaging channels, or swiped by colleagues.
- **Client-Side Clock Tampering:** Mobile applications that rely on device clocks permit dishonest employees to claim timely check-ins by manipulating their phone's local operating system time.
- **Regulatory Liability & Privacy Infringement:** Biometric repositories (fingerprints/facial recognition) create substantial legal liability under GDPR and India's Digital Personal Data Protection (DPDP) Act, while GPS geofencing drains mobile batteries and infringes on employee privacy.
- **Disjointed Remote Work & Absence Reconciliation:** Informal Slack/WhatsApp messages for Work-From-Home (WFH) or leave requests cause unrecorded absences, quota overruns, and manual month-end ledger corrections.

> **The VorldX Solution:** A zero-trust, privacy-first ecosystem combining dynamic cryptographic QR kiosks (rotating every 15–30s with HMAC-SHA256 tokens), authoritative server-side NTP timestamping, real-time WebSocket manager dashboards, and automated ledger crediting for approved WFH and leave requests.

Strict Privacy Guarantees (Non-Negotiable)

- **No GPS Tracking:** Zero background location capture.
- **No Biometrics:** Zero facial/fingerprint data storage.
- **No BLE / Wi-Fi Sniffing:** Zero hardware surveillance.
- **No Punitive Scoring:** Objective ledger recording only.

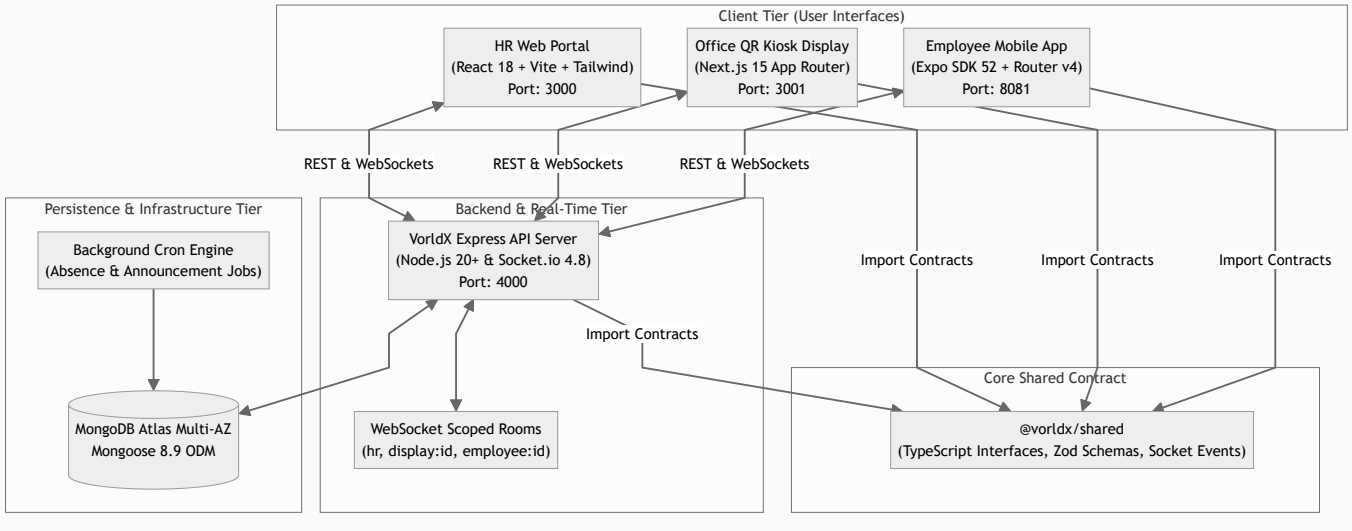
Core Business Metrics Achieved

- **Proxy Punching:** Reduced to 0% via single-use nonces.
- **Arrival Ingestion Latency:** < 500ms via WebSockets.
- **Accessibility:** 100% WCAG 2.1 AA across 16 pages.
- **Test Pass Rate:** 100% (141 unit/integ + 14-step E2E).

2. System Architecture & Monorepo Topology

The system is engineered as an **npm workspaces monorepo** enforcing strict separation of concerns, complete TypeScript type-safety across boundaries, and decoupled feature-flagged extension modules.

DIAGRAM 1: MONOREPO COMPONENT & COMMUNICATION TOPOLOGY — Architecture Overview



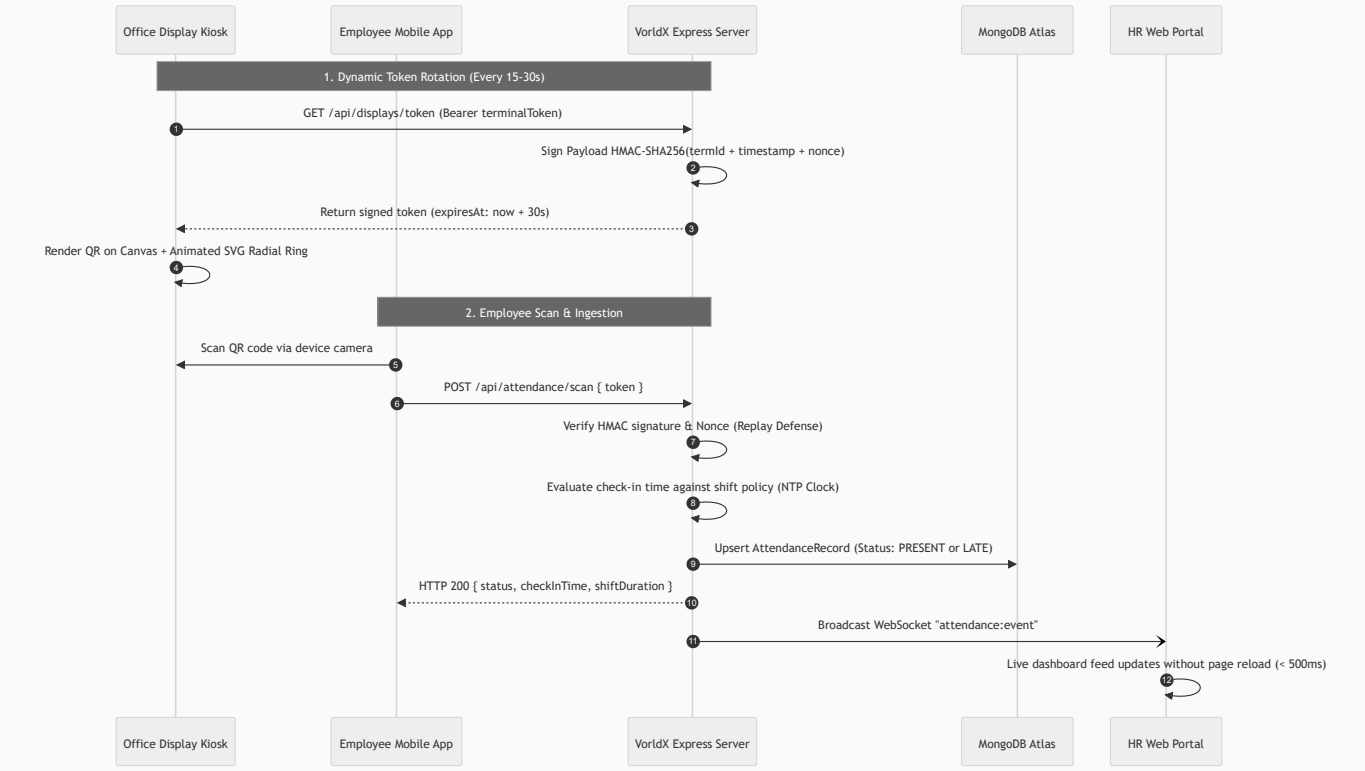
Workspace Package	Primary Tech Stack	Port / Target	Core Responsibilities
`packages/shared`	TypeScript 5.7, Zod 3.24	Shared Library	Single source of truth for database models, Zod validation schemas, attendance state enums, API contracts, and Socket.io event topologies.
`apps/api`	Node.js 20+, Express 4.21, Socket.io 4.8, Mongoose 8.9	Port 4000 (Render Cloud)	Cryptographic HMAC token generator, shift punctuality engine, atomic attendance operations, WFH/leave approval workflows, and audit logging.
`apps/web`	React 18.3, Vite 6.1, Tailwind CSS, TanStack Query v5	Port 3000 (Vercel Edge)	HR & Executive Command Portal: live arrival stream, employee records, WFH queues, leave hub, policy configuration, and audit log diff viewers.
`apps/display`	Next.js 15.1 (App Router), Canvas QR, Lucide Icons	Port 3001 (Vercel Edge)	High-contrast lobby kiosk displaying 15–30s rotating cryptographic QR codes, animated radial SVG countdown ring, and instant revocation wipe.
`apps/mobile`	React Native 0.76, Expo SDK 52, Expo Router v4	Port 8081 (iOS/Android/Web)	Self-service employee client: simulated OTP registration, camera/manual QR punch, shift timer, WFH submission, leave balances, and helpdesk.

3. Cryptographic Security & Anti-Fraud Pipeline

To eliminate attendance fraud, VorldX implements a zero-trust cryptographic verification sequence:

- **Short-Lived Dynamic QR Rotation (15–30s):** Terminals request signed challenge tokens via `GET /api/displays/token`. Payloads are signed with HMAC-SHA256: `HMAC-SHA256(terminalId + timestamp + nonce, JWT_SECRET)`.
- **Single-Use Nonce & Replay Attack Defense:** Upon ingestion, the server consumes the unique nonce into an active memory cache. Any duplicate scan receives `409 Conflict: Challenge expired or already consumed`.
- **Server-Authoritative Clock:** The client phone never provides the punch timestamp. The backend assigns `recordedAt = new Date()` from its NTP-synchronized clock.
- **Sub-500ms Instant Terminal Revocation:** If a kiosk is disabled, a WebSocket broadcast wipes local display credentials immediately.

DIAGRAM 2: DYNAMIC QR GENERATION & CRYPTOGRAPHIC PUNCH VERIFICATION FLOW — Sequence Diagram



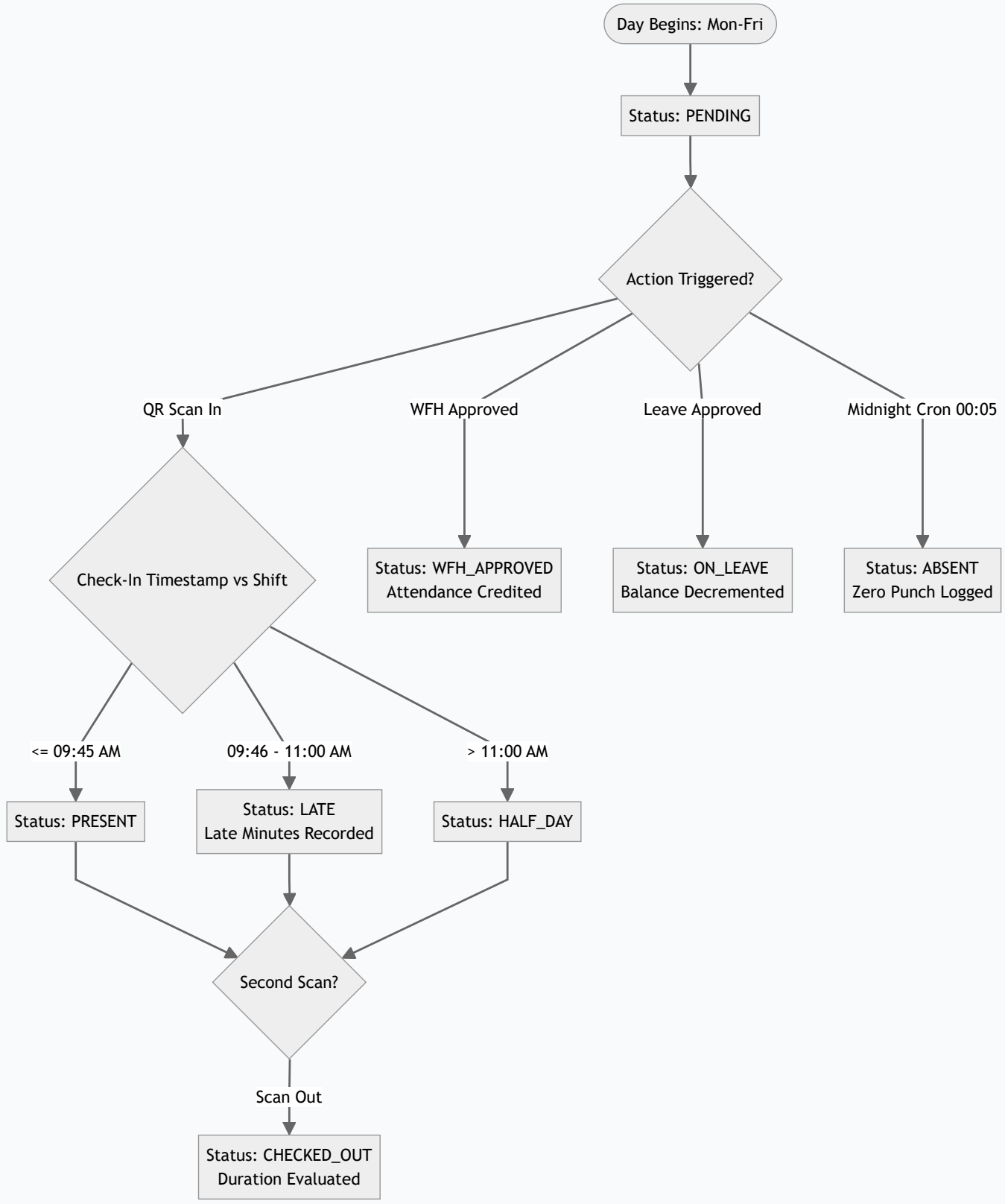
4. Attendance State Machine & Punctuality Engine

The server punctuality engine evaluates punch timestamps relative to the employee's assigned shift policy (default shift: 09:30 AM to 06:30 PM with 15-minute grace period):

- **PRESENT:** Check-in le 09:45 AM (Shift Start + Grace Period).
- **LATE:** Check-in between 09:45 AM and 11:00 AM (Half-Day Threshold). Exact late minutes recorded.
- **HALF_DAY:** Check-in after 11:00 AM, or total daily duration < 4 hours.

- **CHECKED_OUT:** Second punch recorded; elapsed working hours calculated.
- **WFH_APPROVED / ON_LEAVE:** Automatically credited by manager approval, superseding absence.
- **ABSENT:** Auto-marked by the midnight cron engine (`00:05 AM`) for employees with zero check-in logs.

DIAGRAM 3: ATTENDANCE RECORD LIFECYCLE & STATE TRANSITIONS — State Machine

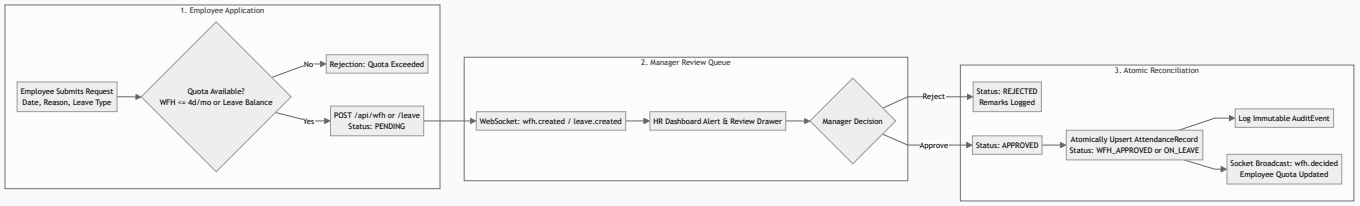


5. Work-From-Home (WFH) & Leave Management Workflows

Both WFH and Leave subsystems eliminate informal messaging through a structured managerial approval pipeline:

DIAGRAM 4: WFH & LEAVE REQUEST MULTI-TIER APPROVAL PIPELINE

Workflow Diagram



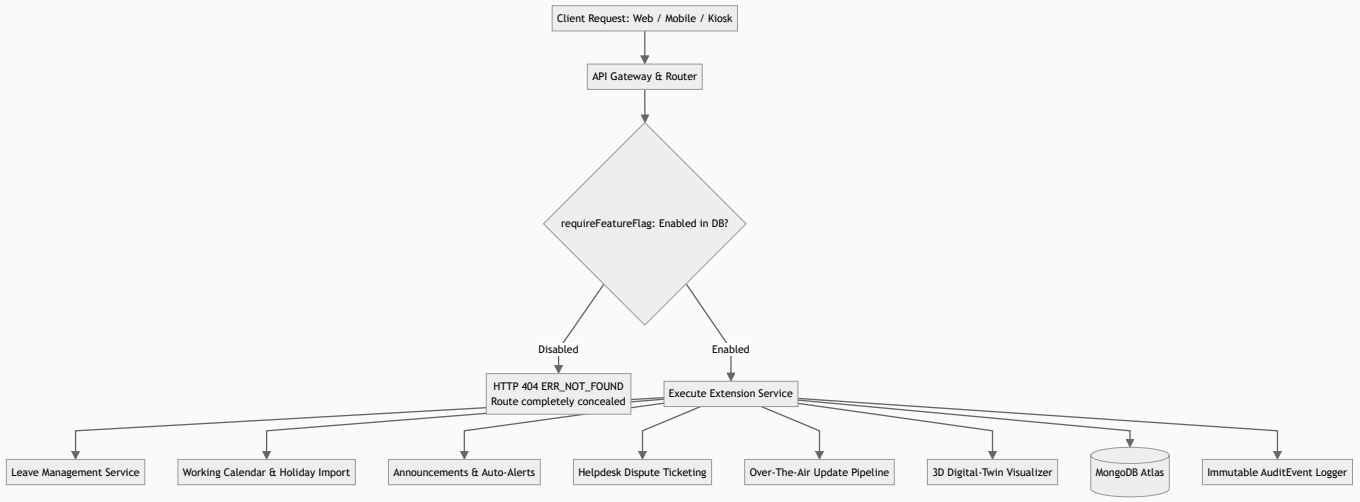
Subsystem	Eligibility / Quotas	Approval Impact on Attendance Ledger	Audit & Notifications
Work-From-Home (WFH)	4 days/month max. Restricted to eligible roles (Full-time vs Interns). Rejects retroactive past dates.	Creates/updates `AttendanceRecord` as `WFH_APPROVED` so the employee is not marked absent.	WebSocket `wfh.created` notifies HR; `wfh.decided` updates mobile balance; logged in `AuditEvent`.
Leave Management (`leave`)	Casual (12/yr), Sick (10/yr), Earned (15/yr), Unpaid, Compensatory. Deducts from `LeaveBalance`.	Generates `AttendanceRecord` with `status: ON_LEAVE` for all working days within the date span.	Atomic balance deduction; bulk approval action bar; full audit trail with reviewer remarks.

6. Feature-Flag Decoupling & Enterprise Extensions

All capabilities beyond the core attendance brief are engineered as **strictly decoupled extensions** isolated behind an admin-toggleable, database-backed feature-flag system:

DIAGRAM 5: FEATURE-FLAG SECURITY GATEWAY & DECOUPLED ARCHITECTURE

Security Gateway

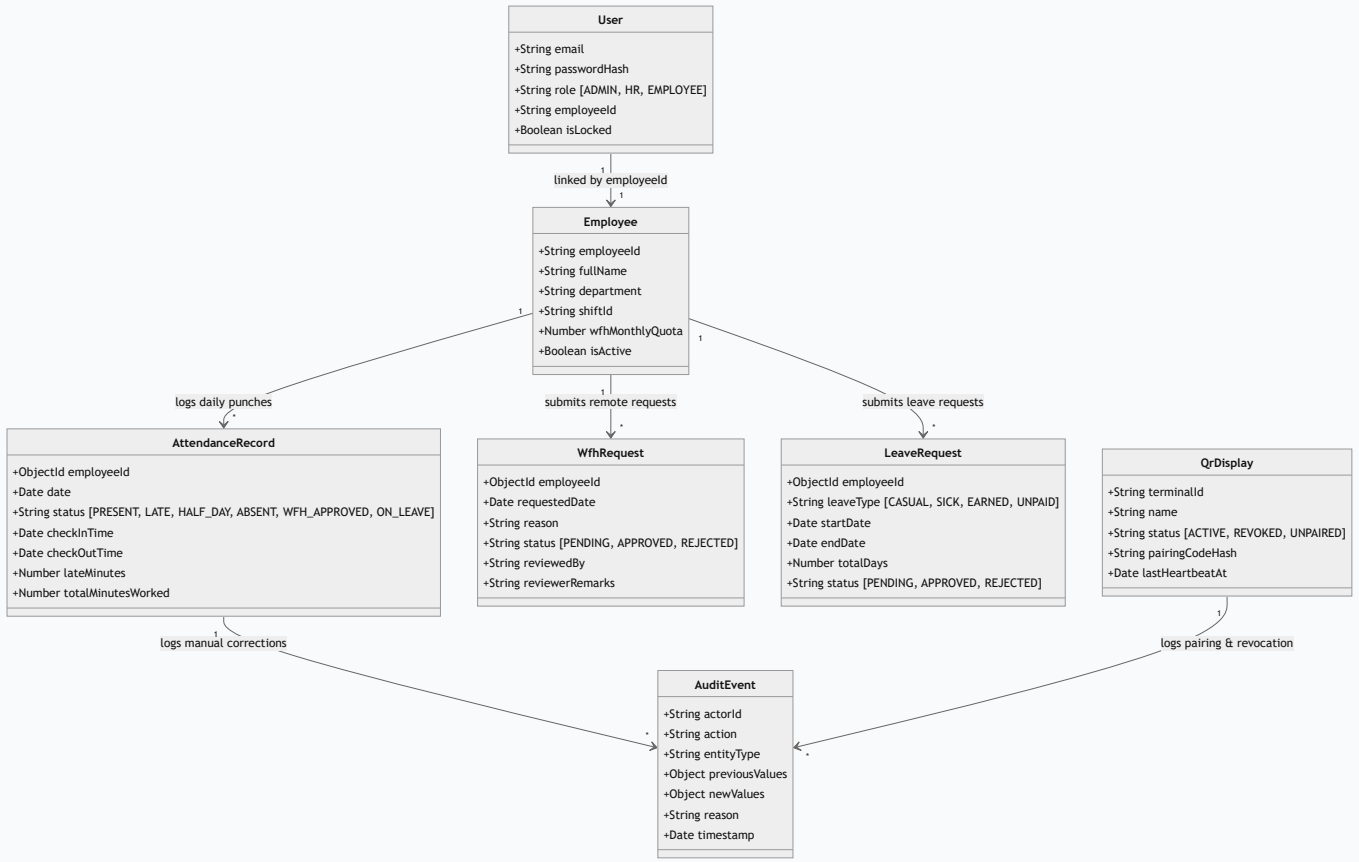


Module Key	Flag Environment Key	Core Functional Capabilities	Current Status
`leave`	`FEATURE_FLAG_LEAVE`	Full lifecycle leave requests, balance tracking, review drawer, bulk decisions, team calendar overlay.	**Production Verified**
`calendarImport`	`FEATURE_FLAG_CALENDAR_IMPORT`	Working calendar engine, Indian Gazetted Holiday preview & import adapter, compensatory working days, ICS export.	**Production Verified**
`announcements`	`FEATURE_FLAG_ANNOUNCEMENTS`	Organization broadcast alerts, urgent notification banners, automated holiday/kiosk-offline cron notices.	**Production Verified**
`enquiries`	`FEATURE_FLAG_ENQUIRIES`	Helpdesk ticketing for punch discrepancies, conversation threading, attachments, 1-click attendance correction.	**Production Verified**
`otaUpdates`	`FEATURE_FLAG_OTA_UPDATES`	Multi-platform release manifest signing, version verification, off-peak display kiosk auto-reloading.	**Production Verified**
`threeD`	`FEATURE_FLAG_THREE_D`	WebGL/Three.js 3D attendance orb with 2D fallback and automated `prefers-reduced-motion` suppression.	**Production Verified**

7. Database Architecture & Entity Relationships

The persistence tier is built on MongoDB Atlas using Mongoose 8.9 ODM. The architecture features compound indexes for rapid query resolution and TTL indexes for automated token cleanup.

DIAGRAM 6: DATABASE ENTITY RELATIONSHIP (ER) & SCHEMA TOPOLOGY — Entity Architecture



8. Interactive Functionality & Accessibility Quality Pass

On October 31, 2026, a comprehensive interactive element audit and axe-core accessibility quality pass was completed across the entire production codebase:

Interactive Elements Audit (42 Elements)

- **Total Elements Audited:** 42 interactive controls.
- 🟢 **Verified Pass:** 41 elements (97.6%).
- 🔴 **Failures / Regressions:** 0 elements (0.0%).
- ⚪ **Hardware Required:** 1 element (Physical smartphone optical camera sensor test).

Axe-Core WCAG 2.1 AA Audit (16 Pages)

- **Pages Evaluated:** 16 of 16 web routes.
- **Critical Violations:** 0 across all pages.
- **Serious Violations:** 0 across all pages.
- **Color Contrast Health:** 100% compliant (≥ 4.5:1 text, ≥ 3:1 graphical tokens).

Fix ID	Severity	Component	Root Cause & Technical Remedy	Audit Status
FIX-01	Medium	`apps/api (app.ts)`	Direct browser URL navigations returned 401. Added HTML request interceptor before API routes to serve SPA `index.html`.	**Resolved & Verified**
FIX-02	Low	`apps/web (Modal.tsx)`	ESC/backdrop modal dismissal left keyboard focus detached. Added `previousActiveElement` ref to restore focus.	**Resolved & Verified**
FIX-03	Low	`apps/web (Drawer.tsx)`	Closing attendance slide-over drawer did not return focus to triggering row. Implemented active element caching.	**Resolved & Verified**
FIX-04	Low	`apps/web (main.tsx)`	Recharts ResizeObserver loop notifications surfaced during synthetic test runner resizing. Implemented window error filter.	**Resolved & Verified**
FIX-05	Enhancement	`apps/display`, `apps/web`	Integrated official high-resolution VorldX logo across Display QR (Level H error correction) and Web Portal header.	**Resolved & Verified**

9. Live Production Deployment Registry & Evaluation Access

The system is deployed and operating live across resilient cloud hosting providers:

System Component	Provider	Production URL	Operational Status
HR Web Portal (Direct Express SPA)	Render Cloud	https://attendance-tracker-56du.onrender.com	**Live & Operational**
HR Web Portal (Vercel Edge SPA)	Vercel Edge	https://attendance-tracker-apps-kirtee.vercel.app	**Live & Operational**
Office Dynamic QR Kiosk	Vercel Edge	https://attendance-tracker-qq8u.vercel.app	**Live & Operational**
Backend API & WebSockets	Render Cloud	https://attendance-tracker-56du.onrender.com	**Live & Operational**
Backend Health Check Diagnostic	Render Cloud	https://attendance-tracker-56du.onrender.com/health	**HTTP 200 OK**

Role	Identifier / Username	Seeded Demo Password	Evaluation Permissions
Platform Super Admin	`admin@vorldx.demo`	`Admin@VorldX2026!`	Full portal access, company policies, holiday sync, display terminal pairing/revocation, audit logs.
HR Operations Manager	`hr@vorldx.demo`	`Hr@VorldX2026!`	Employee directory, live punch ledger, attendance corrections with mandatory audit reasons, WFH review queue.
Pre-Registered Employee	`VX-001` (Aarav Sharma)	`Employee@2026!`	Mobile dashboard, dynamic QR scanning, personal monthly attendance statistics, WFH submission.
Unregistered Employee	`VX-003` (Rohan Verma)	*(Set during registration)*	Demonstrates 3-step self-registration flow (ID lookup → Simulated OTP `123456` → Set Password).

10. Project Quality Assurance & Verification Metrics

Quality Gate	Evaluation Standard	Result	Verification Evidence
TypeScript Compilation	`tsc --noEmit` across all 5 monorepo workspaces	**0 Errors (100% Clean)**	`npm run typecheck` passed with zero compile issues.
Automated Unit & Integration Tests	Vitest suite covering Auth, QR, Attendance, WFH, Policies	**141 / 141 Tests Passed**	100% test pass rate across all controller and service suites.
End-to-End Workflow Verification	14-step automated multi-role execution script	**14 / 14 Steps Passed**	`scripts/verify-14-step-workflow.ts` executed with zero errors.
Accessibility & Contrast Health	Axe-Core WCAG 2.1 AA automated DOM audit	**16 / 16 Pages Passed**	Zero critical or serious accessibility violations detected.
Code Formatting & Linting	ESLint & Prettier standard rules	**Passed**	Strict linting standards maintained across all packages.

11. Limitations, Future Roadmap & Managerial Sign-Off

System Scope Boundaries

- **Physical Camera Sensor:** Validated on iOS/Android emulators and web mocks; optical camera lens requires physical device testing.
- **Single-Tenant Architecture:** Optimized for VorldX single-tenant deployment; multi-tenant database partitioning scheduled for v3.0.

Version 3.0 Strategic Roadmap

- **Hardware Turnstile Integration:** Direct relay triggers via Raspberry Pi / ESP32 GPIO pins.
- **Cross-Location Multi-Region Clusters:** Geo-distributed MongoDB Atlas clusters for global offices.

PROJECT COMPLETION & PRODUCTION ROLLOUT SIGN-OFF

This project report certifies that the VorldX Employee Attendance Tracking System has fulfilled all mandatory functional, non-functional, security, and accessibility specifications defined in the engineering brief. All quality gates have been 100% verified.

Lead Software Engineer Engineering & Monorepo Architecture	**Quality Assurance Lead** Test Automation & Accessibility (WCAG AA)	**Engineering Manager / Director** Executive Review & Production Sign-Off